National University – Sudan
Faculty Of Engineering & Architectur
Academic year 2023/2024 – semester1
Med Exam

1/ Solve the following matrix equation for a, b, c, and d

a/ $\begin{bmatrix} a+2b & 2a-b \\ 2c+d & c-2d \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 4 & -3 \end{bmatrix}$

b/ $\begin{bmatrix} 2b & -2 \\ 2 & 2a \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 2 & -8 \end{bmatrix}$

2/ Consider the matrix:

$A = \begin{bmatrix} 3 & -2 \\ 2 & 4 \end{bmatrix}$, B = $\begin{bmatrix} -4 & 5 \\ 2 & 3 \end{bmatrix}$, C = $\begin{bmatrix} 1 & 5 \\ 4 & 0 \\ -1 & 7 \end{bmatrix}$

Find :

1/ A+ B , 2/ A – B , 3/ AB , 4/ BA , 5/ 3A – 2B

6/ 5B , 7/ A^2 , 8/ B^2 , 9/AC , 10/CB

3/Find the Inverse (A^{-1})

$$\begin{bmatrix} 4 & -1 \\ 3 & -2 \end{bmatrix} \quad , \quad \begin{bmatrix} 2 & 2 \\ 3 & -1 \end{bmatrix}$$

4/Find the determinant det(A)

$$\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix} \quad , \quad \begin{bmatrix} -1 & -3 \\ 6 & 2 \end{bmatrix} \quad , \quad \begin{bmatrix} 4 & 2 & 1 \\ 5 & 2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

4/ Composition of two matrix:

Find : f0g , g0f , f0g(3) , gof (-1)

a- F(x) = 3x -4 , g(x) = 5x + 2

b- F(X) = X2 , g(x) = 3x

c- F(x) = 2x , g(x) = x +1

d-

5/ find the slope of line that path through two point (
1,3), (2,5)

6/ Find the slope of 5y = 10 x + 5

7/ find the equation of the line that has a slope of m = 5 and passes through the point (2,2)

8/ let A = $\begin{bmatrix} 4 & 5 & 0 \\ -1 & -3 & 3 \\ 5 & 6 & 7 \end{bmatrix}$

Evaluate det(A) by cofactor

Good Luck

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$a + 2b$